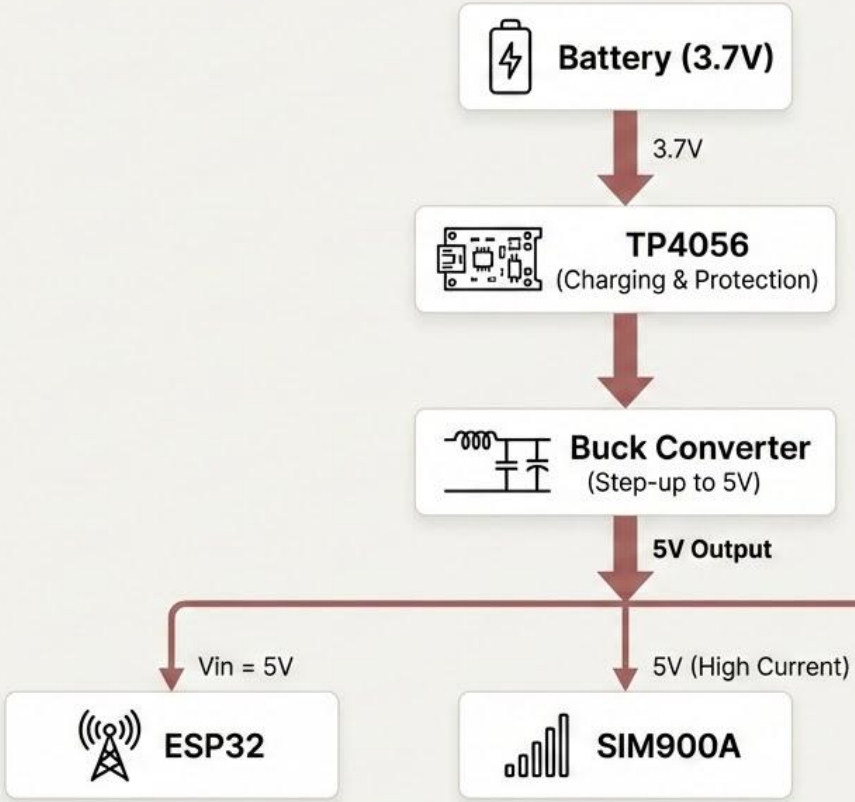
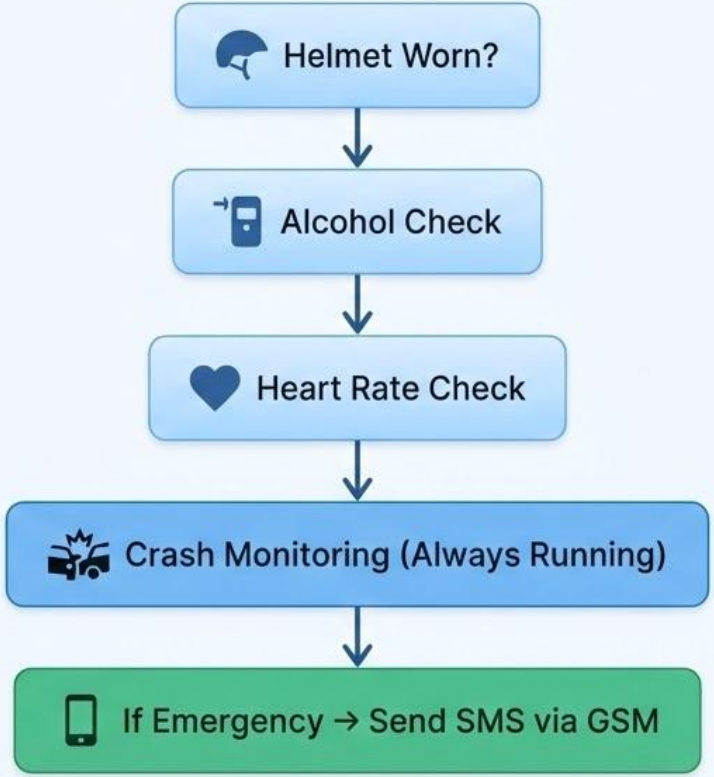
Power Supply & Component Integration



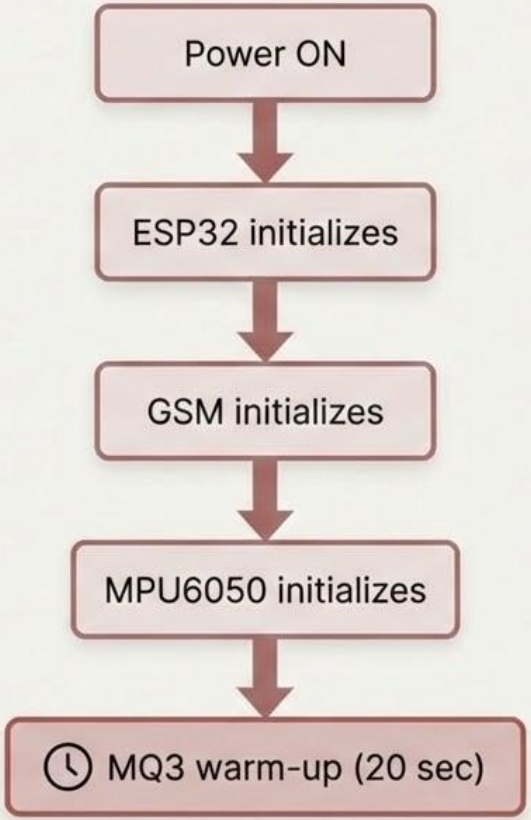
Important Warnings:

- Buck Converter Output = 5V
- ESP32 Vin = 5V
- SIM900A = 5V (High Current)
- MPU6050 + Pulse sensors derive 3.3V directly from ESP32's 3.3V output pin

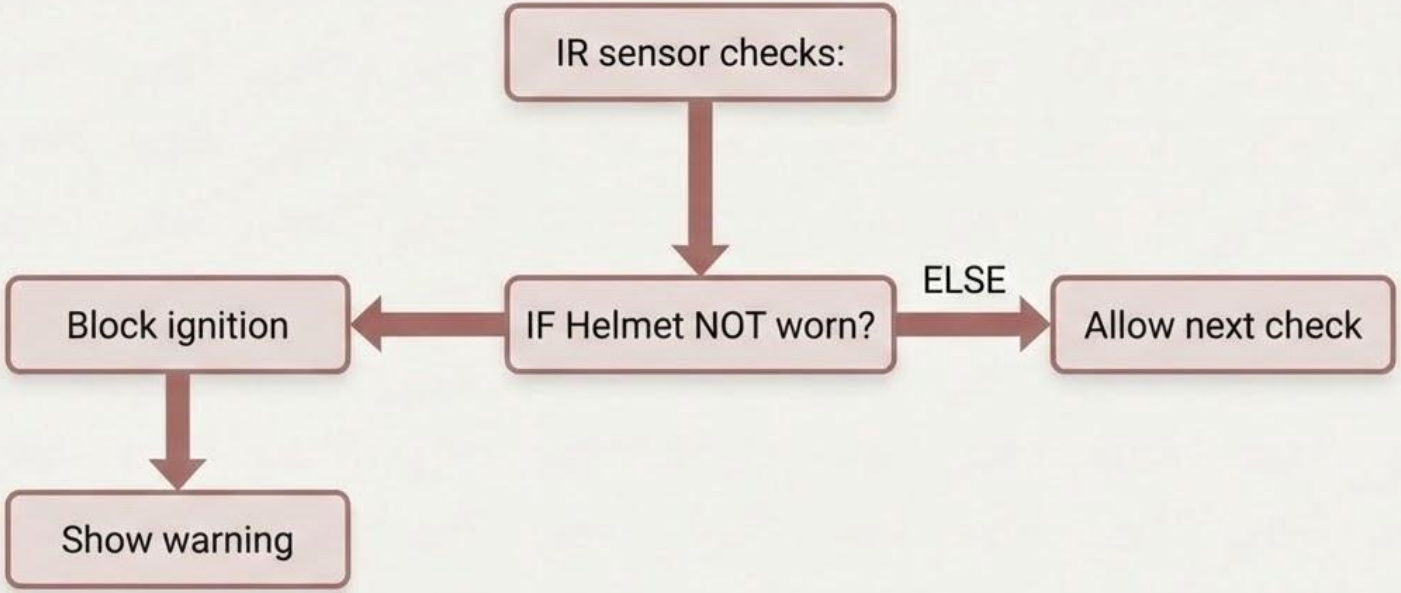
Smart Helmet Safety System Workflow



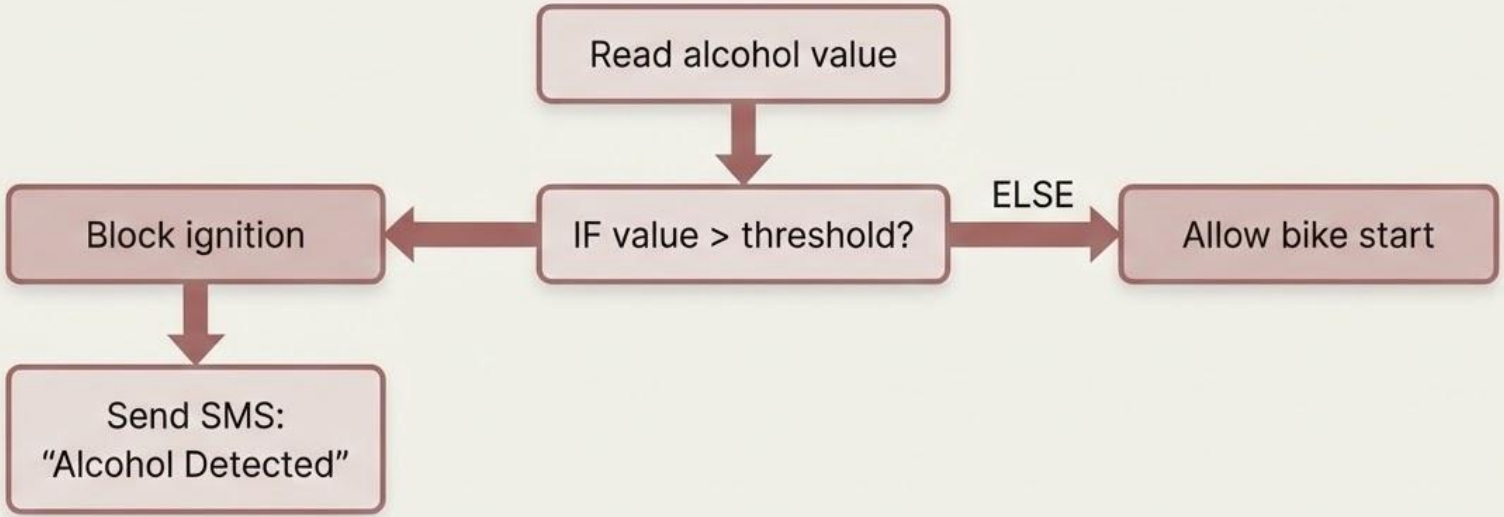
STEP 1 — System Start



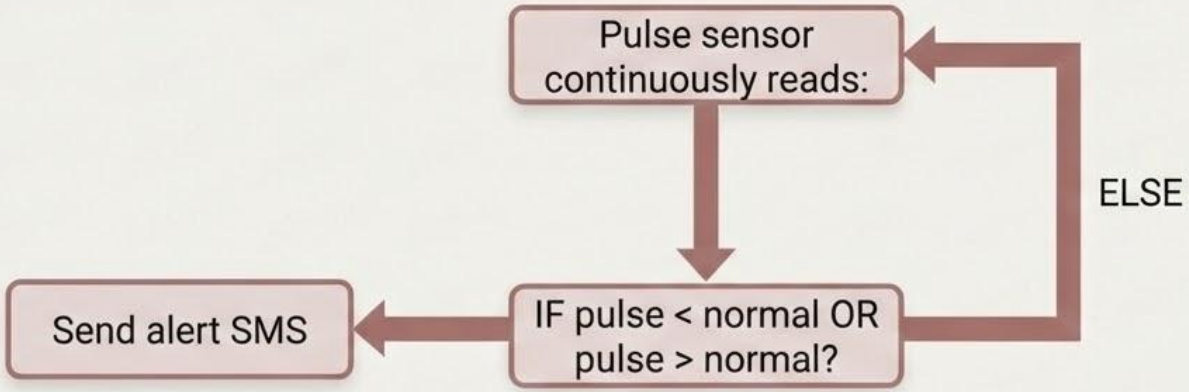
STEP 2 — Helmet Detection



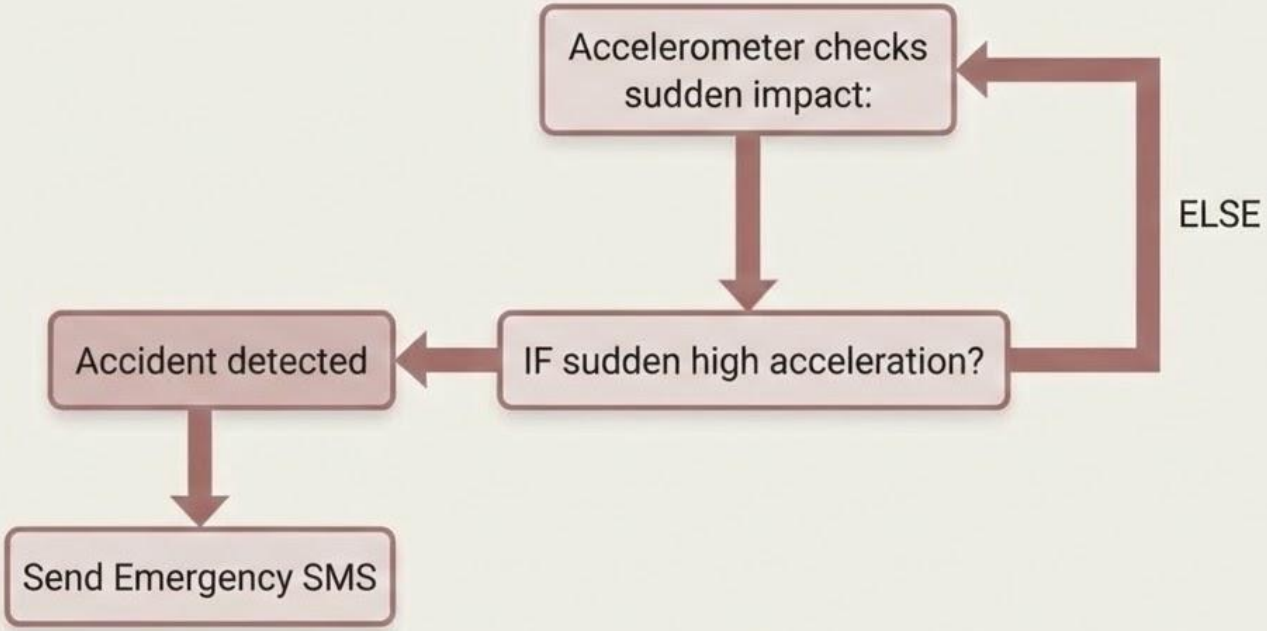
STEP 3 — Alcohol Detection (MQ3)



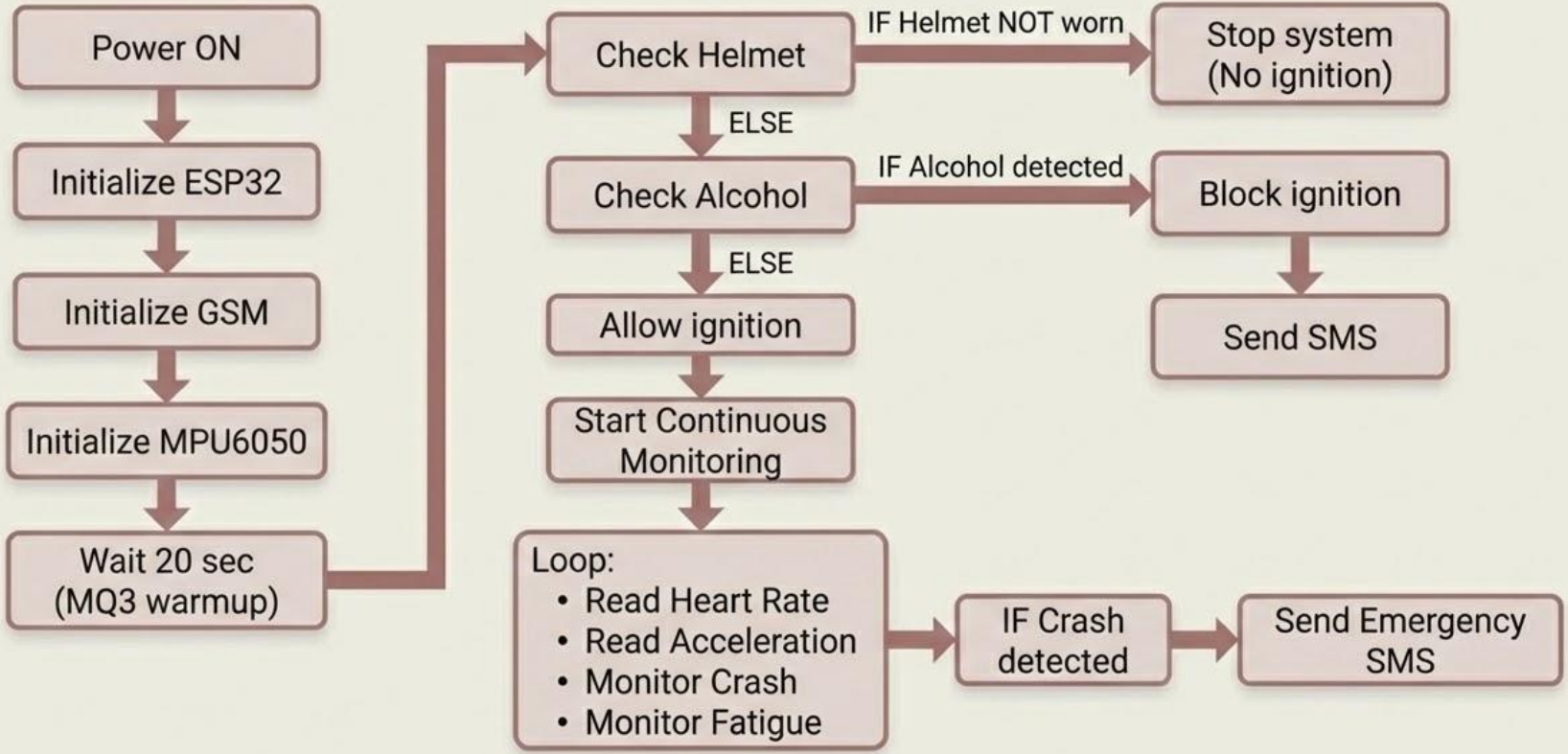
STEP 4 — Heart Rate Monitoring



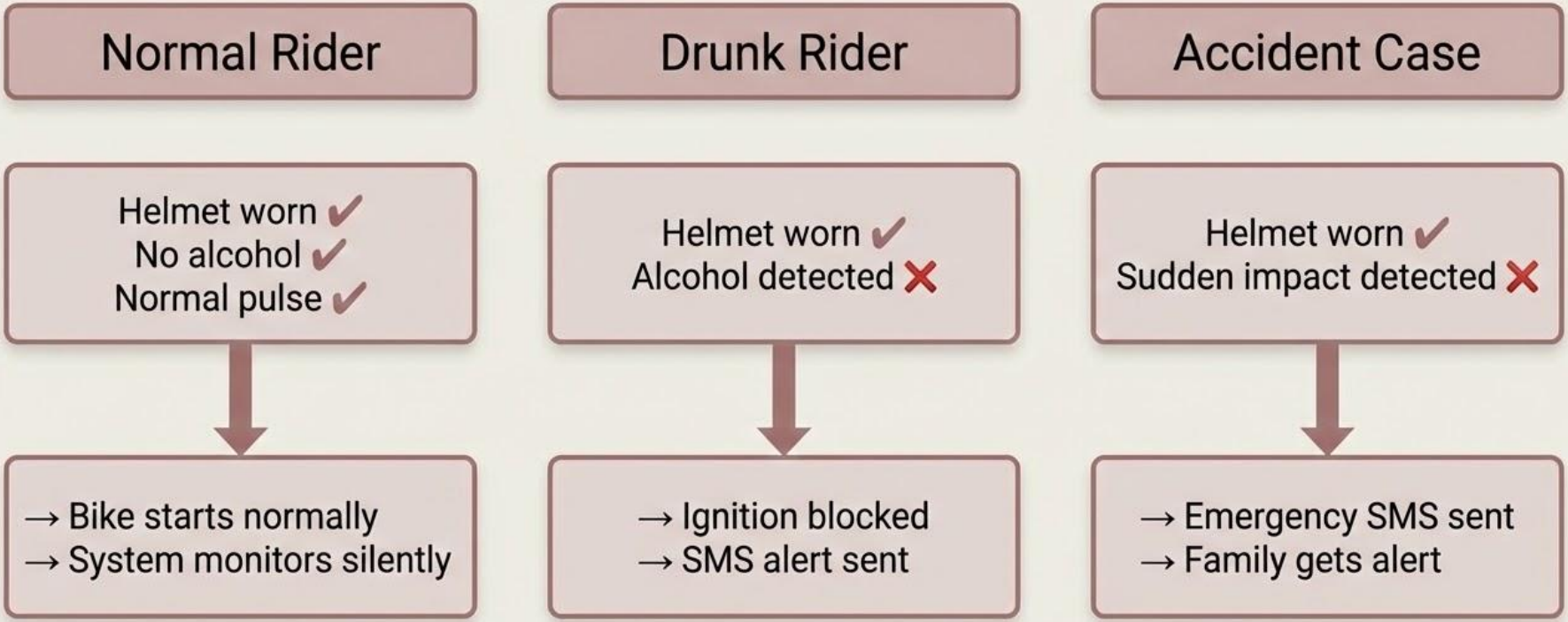
STEP 5 — Crash Detection (MPU6050)



Complete Working Logic

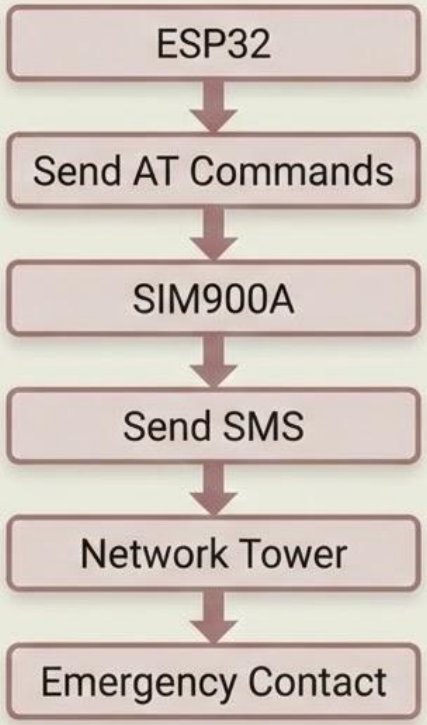


REAL WORLD SCENARIO FLOW

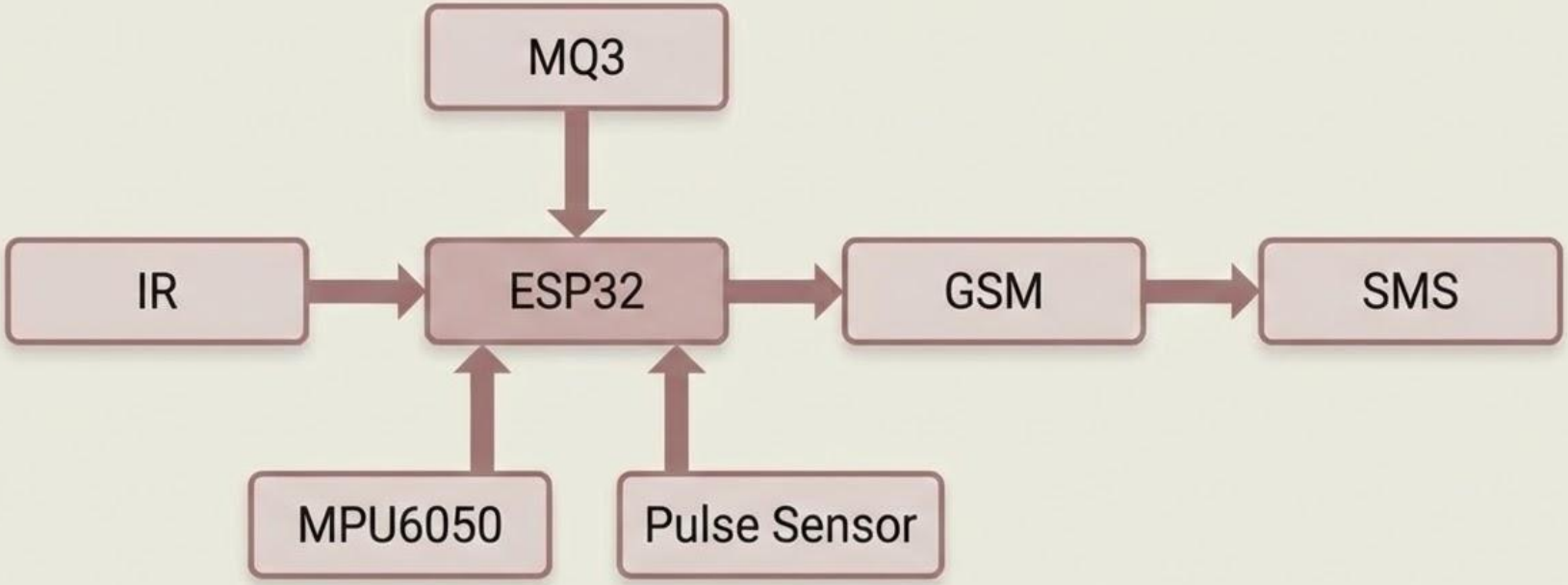


GSM FLOW

When emergency detected:



MODULE RELATION FLOW



ESP32 = Brain Sensors = Inputs GSM = Communication